

YUHUA ZHANG

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EDUCATION

University of Michigan, Ann Arbor, MI Ph.D. in Biostatistics Supervised by Dr. Walter Dempsey & Dr. Sebastian Zöllner	2019.9 - 2023.8
University of Michigan, Ann Arbor, MI M.S. in Biostatistics	2017.9 - 2019.4
Shanghai Jiao Tong University, Shanghai, China B.S. in Biological Science	2013.9 - 2017.6

EXPERIENCE

Yale University, New Haven, CT Postdoctoral Researcher Supervised by Dr. Shuangge Ma	2026.8 - Present
Harvard University, Boston, MA Postdoctoral Researcher Supervised by Dr. Jukka-Pekka Onnela	2023.9 - 2026.7

RESEARCH INTERESTS

Network Data Analysis, Causal Inference with Network Interference, Healthcare Delivery in Elder Population, Large-scale Claim Data Analysis, Bayesian Nonparametric Methods, Community Detection

PUBLICATIONS AND PREPRINTS

Theory and Statistical Methodology

- Yuhua Zhang**, JP Onnela, Shuo Sun, Ruoyu Wang. “Identification and estimation of heterogeneous interference effects under unknown network.” *ArXiv preprint*, (2025). Available at arXiv.
- Yuhua Zhang**, Kori S. Zachrison, Renee Y. Hsia, JP Onnela. “Community detection through recursive partitioning in Bayesian framework.” *ArXiv preprint*, (2025) (Under revision at *Biometrics*).
- Emma Crenshaw, **Yuhua Zhang**, and JP Onnela. “Covariate selection for joint latent space modeling of sparse network data.” *ArXiv preprint*, (2026) (Under review at *JRSSB*).
- Yuhua Zhang**, and Walter Dempsey. “Node-level community detection within edge exchangeable models for interaction processes.” *Journal of the American Statistical Association* 120.550 (2025): 764-778.
- Yuhua Zhang**, and Walter Dempsey. “CataBEEM: integrating latent interaction categories in node-wise community detection models for network data.” *Proceedings of the 40th International Conference on Machine Learning (ICML)*, PMLR 202:40946-40975, 2023.
- Yuhua Zhang**, Corbin Quick, Ketian Yu, Alvaro Barbeira, GTEx Consortium, Francesca Luca, Roger Pique-Regi, Hae Kyung Im, and Xiaoquan Wen. “PTWAS: investigating tissue-relevant causal molecular mechanisms of complex traits using probabilistic TWAS analysis.” *Genome Biology* 21, no. 1 (2020): 232.

Interdisciplinary Application

7. Kori Zachrison, **Yuhua Zhang**, Mathew Reeves, Renee Hsia, Yizhou Cui, Lee Schwamm, Vicki Fung, Margaret Samuels-Kalow, Carlos Camargo, Jr., and Jukka-Pekka Onnela. “Changes in stroke care regionalization in California associated with evolving evidence of benefit for endovascular thrombectomy: a network analysis.” (Under revision at *Circulation: Population Health and Outcomes*)
8. Karadakic, René, David C. Chan, Bruce E. Landon, Nancy L. Keating, Christopher Manz, Jukka-Pekka Onnela, Thomas C. Tsai, **Yuhua Zhang**, and Michael L. Barnett. “Subspecialization of surgical specialties in the US.” *JAMA Health Forum*. Vol. 6. No. 9. American Medical Association, (2025).
9. Ryan, Kelly A., Anastasia K. Yocum, **Yuhua Zhang**, Peisong Han, David F. Marshall, Paul T. Costa, Sarah H. Sperry, Takakuni Suzuki, Melvin G. McInnis, and Sebastian Zöllner. “Predictive evidence for the impact of personality styles on depression and functioning in two bipolar disorder cohorts.” *Journal of Affective Disorders* 380 (2025): 746-755.
10. Ryan, Kelly A., Peisong Han, **Yuhua Zhang**, David F. Marshall, Anastasia K. Yocum, Melvin G. McInnis, and Sebastian Zöllner. “Stability of personality traits in bipolar disorder: findings from a longitudinal cohort.” *Journal of Affective Disorders* 283 (2021): 1-10.
11. Hu, Yiming, Qiongshi Lu, Wei Liu, **Yuhua Zhang**, Mo Li, and Hongyu Zhao. “Joint modeling of genetically correlated diseases and functional annotations increases accuracy of polygenic risk prediction.” *PLoS Genetics* 13, no. 6 (2017): e1006836.

SELECTED MANUSCRIPTS IN PREPARATION

12. **Yuhua Zhang**, Walter Dempsey. “Conformal link prediction in edge exchangeable model.” *Manuscript available upon request* (2026+).
13. **Yuhua Zhang**, Michael L. Barnett, JP Onnela. “Bayesian hierarchical community detection method of network data.” *Manuscript available upon request* (2026+).

FUNDING EXPERIENCE

NIH K99/R00 Pathway to Independence Award

Submitted Oct 2025; Not funded

Title: Quantifying Network Effects of Reperfusion Interventions in Stroke Systems of Care

Role: PD/PI

PRESENTATIONS

Invited Talk

1. Department of Statistics & Data Science, Carnegie Mellon University, “From communities to causes: statistical methods for network data”, Mar 2026.
2. Department of Epidemiology and Biostatistics, University of Arizona, “From communities to causes: statistical and ML methods for network data”, July 2026.

Conferences

3. Joint Statistical Meeting, “Causal inference with the presence of latent interference network”, Aug 2025.
4. Joint Statistical Meeting, “Identification of subspecialty among physicians – community detection in Medicare claim data”, Aug 2024.
5. Western North American Region Spring Meeting, “Conformal link prediction”, Mar 2024.
6. Eastern North American Region Spring Meeting, “Community detection within edge exchangeable models for interaction processes”, Mar 2023.

TEACHING INTERESTS

Undergraduate

- Probability & Statistics (core course for stats/biostats majors)
- Regression & ANOVA (linear models with R/Python labs)
- Introduction to Data Science (data preprocessing, EDA, visualization, reproducibility)

Graduate

- Bayesian Inference (conjugate priors, MCMC, hierarchical models; hands-on experience with Stan)
- Statistical Learning (regularization, trees/boosting, cross-validation, neural network; applied projects)
- Advanced Elective: Theory and Application of Network Analysis (network models, community detection, network-assisted inference; case studies)

ENGAGEMENTS & SERVICES

Department of Biostatistics, University of Michigan, Ann Arbor

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| • Departmental Search Committee student representative | 2021.10 - 2022.4 |
| • Co-founder and member of the Student Research Conflict Committee | 2020.8 - 2021.4 |
| • Departmental DEI Committee member | 2019.1 - 2020.6 |

COMPUTING SKILLS

Computing language: Python, R, SQL

[Compiled on July, 2026]